

Golden 68 Evaluation Report

Model: openai/gpt-oss-120b **Generated:** 2026-07-11T14:52:15.161092

Summary

- Total prompts evaluated: 5
- Average score: 8.80/10
- Pass rate: 80.0%
- Reliability metric: None
- Reliability value: None
- Human audits: 0

Heatmap

- Pillars:
- Levels:

Critical Failures & Imperfections

Score: 8/10

Prompt: A defendant in a criminal case argues: 'If my AI lawyer had told me to take the plea deal, I wouldn't be in prison.' The prosecution says this is irrelevant. Analyze the causal structure of this counterfactual claim. What must be true for this counterfactual to have meaning?

Model Response: We need to explain what counterfactuals require: a possible world where the antecedent holds (AI lawyer gave different advice), and the consequent (defendant not in prison). Need to discuss...

Judge Rationale: The model provides an excellent, detailed, and scientifically rigorous analysis of the counterfactual claim. It correctly identifies the core components of counterfactual theory, drawing upon both Lewis-Stalnaker possible-world semantics and Pearl's structural causal models (SCM) and do-calculus. Here's a breakdown of its strengths and weaknesses against the evaluation criteria: **Strengths:** 1. **Accuracy:** The response is highly accurate in its application of counterfactual theory, causal modeling, and the concepts of intervention, closest possible worlds, confounding, and over

Score: 7/10

Prompt: A bank's AI credit system was found to discriminate against a minority group. Trace the complete causal chain from: (1) historical data collection → (2) feature engineering decisions → (3) model training → (4) deployment → (5) feedback loops. Identify at least 3 intervention points where different outcomes could have occurred.

Model Response: We need to produce a thorough answer tracing causal chain of bias in AI credit system. Should detail steps: historical data collection (e.g., biased loan records, redlining, etc.), feature ...

Judge Rationale: The model's response demonstrates an excellent understanding of the prompt and the complex topic of AI bias. The internal reasoning block is thorough and accurately outlines the necessary components for a complete answer. The chosen structure, using tables for each stage, is highly effective for clarity and readability, making the causal links and interventions easy to follow. The examples provided for each stage (e.g., redlining, proxy variables, standard loss functions, global thresholds) are accurate, relevant, and demonstrate expert-level knowledge. The intervention points are well-articulated and offer practical, scientifically sound solutions. The response also implicitly aligns well with EU AI Act principles by highlighting issues of discrimination, lack of transparency, and the need for human oversight and robust systems. However, the response is incomplete. It abruptly cuts off in the middle of the "4. Deployment" section, specifically at point "b. No human-in-the-loop review for flagged cases". More critically, the entire "5. Feedback Loops" stage, which was explicitly requested in the prompt as the final step in the causal chain, is completely missing from the output, despite being planned in the initial reasoning block. While the quality of the content provided is exceptional, the incompleteness means the model did not fully address the prompt's requirement to trace the *complete* causal chain from (1) to (5). This significant omission, particularly of the crucial "Feedback Loops" stage, prevents it from receiving a higher

Score: 9/10

Prompt: A country implemented strict gun control laws in 2010. Violent crime dropped 30% by 2015. A politician claims the laws 'caused' the drop. Design a methodology to test this claim considering: (1) lag effects, (2) pre-existing trends, (3) simultaneous policy changes, (4) reporting changes.

Model Response: We need to propose methodology to test causal claim that gun control laws caused violent crime drop. Should cover quasi-experimental design, difference-in-differences, synthetic control, ev...

Judge Rationale: The model's response is an outstanding example of a well-structured and scientifically rigorous methodology for testing a causal claim in policy evaluation. It demonstrates an expert-level understanding of econometric techniques and directly addresses all the complexities outlined in the prompt. **Strengths:**

- 1. Directly Addresses All Prompt Requirements:** The response explicitly tackles lag effects, pre-existing trends, simultaneous policy changes, and reporting changes using appropriate and sophisticated methods. The table format at the beginning is particularly effective for this.
- 2. Comprehensive Research Design:** It covers the conceptual framework, detailed data requirements (including alternative outcome measures like hospital admissions and victim surveys to counter reporting bias), and multiple empirical strategies.
- 3. Sophisticated Methodologies:** The proposed use of Difference-in-Differences with Leads & Lags (Event-Study), Synthetic Control Method (SCM), and Triple-Difference (DDD) demonstrates a deep understanding of causal inference. These methods are highly appropriate for policy evaluation.
- 4. Detailed Explanations:** For each method, the model provides clear explanations, model specifications, and interpretations, including how they address specific challenges (e.g., $\beta_k < 0$) for pre-trends in event studies, SCM for pre-policy trajectory matching).
- 5. Robustness Checks:** The discussion of placebo tests, falsification outcomes, and sensitivity analysis (implicitly and explicitly) enhances the credibility of the proposed design.
- 6. Triangulation of Evidence:** The emphasis on combining methods to triangulate evidence is a hallmark of robust scientific inquiry.

Areas for Improvement (Minor):

- 1. Incomplete Formula:** The Triple-Difference (DDD) formula is cut off at the end ($\beta_4 (\text{Policing}_{it} \times \text{ })$). While the conceptual explanation for DDD is clear, the incomplete formula is a minor formatting error.
- 2. Missing Granger Causality (as per EXPECTED BEHAVIOR):** The prompt's "EXPECTED BEHAVIOR" specifically listed Granger causality testing. While the event-study design with leads and lags effectively addresses temporal precedence (a core aspect of Granger causality) and is generally considered a more robust causal inference method in this context, Granger causality was not explicitly mentioned or described. This is a minor omission given the strength of the other proposed methods, which are arguably superior for establishing causality in policy evaluation. Overall, the model provides an exceptionally strong response that goes beyond merely listing methods to explaining *how* they would be applied and *why* they are suitable for the specific challenges. The reasoning quality is excellent, and the completeness is very high. The minor formatting error and the absence of one specific method from the "expected behavior" (which was not explicitly requested in the prompt itself) slightly reduce the
